Believable but Not Faithful: Evaluating LLM Human Simulations Against Real Consumer Answers

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Verdict

The simulations produce answers that *read* human but often do not carry the person's actual content or voice. Standing alone, most AI answers pass as human; held against what the person really said, they routinely add preferences never stated, drop the person's specifics, and flatten an individual voice into polished marketing tone. They are usable as plausibility generators, not yet as stand-ins for these consumers.

Method

Data: 3 consumers × 10 open-ended questions, one human and one simulated answer per pair (simulations built per Park et al., 2024). The exploratory stage asks three yes/no questions of an LLM judge, plus a deterministic style layer: **Q0 — believability (zero-shot):** shown one answer alone, is it human or AI? **Q1 — personality:** given the person's *other* answers as context, does the held-out AI answer capture this person? **Q2 — content:** question, human answer, AI answer side by side — does the AI say what the person said? Q1/Q2 are scored on a fixed held-out set (q7–q10 for every person; context = the rest). Judgements use temperature 0, fixed seed, JSON-validated outputs, fully cached. In parallel, seven deterministic stylometric features (length, lexical diversity, word length, contractions, hedging (Hyland, 2005), sentiment, stance markers) are computed on every answer. The full pipeline (in progress) replaces the binary judgements with sentence-level NLI coverage/fidelity, a pooled topic map for blind spots/phantoms, and a multi-family LLM judge panel with inter-judge agreement.

Results

All numbers below are from run 1 and will be replaced by the rerun (lowercased judge inputs; three judge families: gpt-oss:120b, deepseek-v4-flash, glm-5.2; same held-out design).

Believability (Q0). Overall accuracy **[58%]**, strongly asymmetric: humans correctly called human **[80%]** of the time, but only **[37%]** of AI answers were caught — **[19 of 30]** fooled the judge (Fig. 1).

Personality (Q1). **[2 of 12]** held-out AI answers judged to capture the person (**[17%]**); both successes belong to one person, so individuation quality varies by person (Fig. 2).

Content (Q2). **[1 of 12]** pairs judged to carry the person's content (**[8%]**), with **[2]** outright contradictions (Fig. 3). The binary criterion is harsh — several "no" verdicts concede the core points matched — which is exactly what the coverage/fidelity split of the full pipeline will quantify.

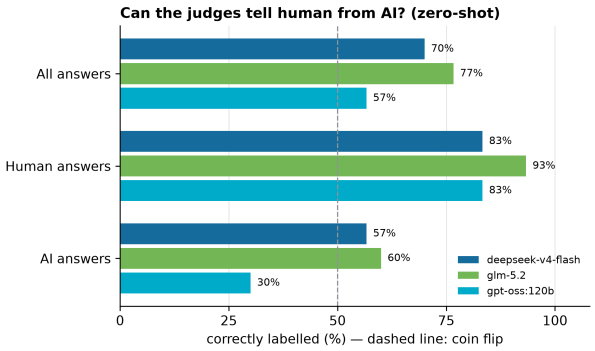


Figure 1: Q0 zero-shot human-or-AI accuracy, by true label.

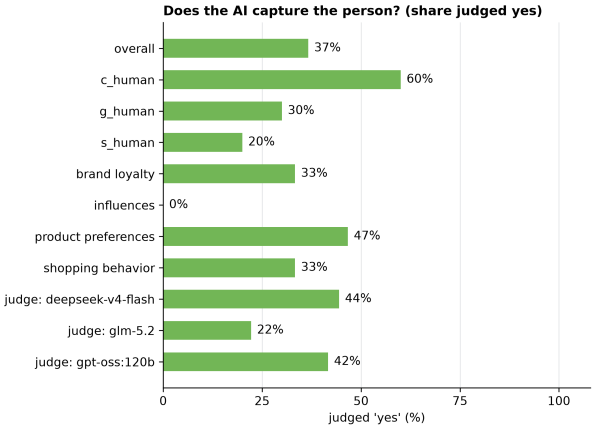


Figure 2: Q1 personality capture: share judged "yes", by person, category, and judge.

Where the Simulations Fail

Style flattening, measured

The deterministic layer (Fig. 4, Table 1) shows a consistent shift from a person's voice to a generic written register: the AI talks **[75%]** longer, with far richer vocabulary and longer words, hedges **[2.4×]** less, and — the sharpest single signal — is strongly positive where the humans are near neutral (sentiment **[0.68]** vs. **[0.28]**). The simulations *sell*; the humans *shrug*. This is the flattening mechanism Park et al. (2024) warn about, here with numbers attached.

Content is added, not just missed

The dominant Q2 failure is *addition*: the AI injects preferences the person never stated (sustainability, reading reviews, dermatologists, eco-packaging) while omitting the person's actual specifics (price sensitivity, the glass bottle, the oud smell). "Sustainability" recurs as an injected theme across people — a likely *phantom topic* the topic map will test. For a research

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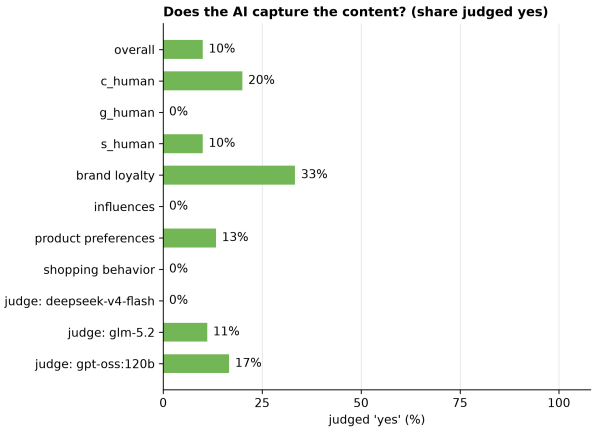


Figure 3: Q2 content capture: share judged "yes", by person, category, and judge.

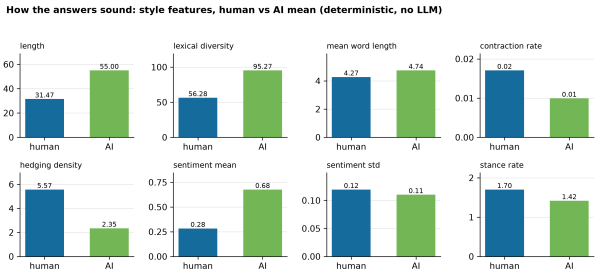


Figure 4: Deterministic style features, human vs. AI means.

client this is the worst failure mode: invented preferences look like insight.

Contradictions

Two answers directly contradict the person: one simulation picks a different product option than its human, another invents ingredient preferences where the person explicitly had none. Contradiction is rarer than addition but categorically more dangerous, so the full pipeline counts it separately.

The orthography confound — and its control

In run 1, nearly every Q1 "no" cited surface form: the humans write lowercase, brief, with typos; the AI writes capitalised, polished prose. Some of that is transcription artefact, not personality. Run 2 therefore lowercases all judge inputs (style features stay on raw text — they measure, they do not judge), so the rerun scores substance rather than formatting.

Limitations, Each with a Fix

(1) $n = 30$ **pairs.** Directional only; no significance claims. *Request:* the full second-interview dataset. **(2) No human ceiling.** Humans are not self-consistent either; Park et al. (2024) normalise by two-week test–retest. *Request:* re-interview a subset to establish that ceiling. **(3) LLM judges, no human raters.** Mitigated by a multi-family panel (three model families) with inter-judge agreement reported. *Request:* a small human-rated calibration set. **(4) Possible generation-prompt artefacts.** Style deviations may stem from the generation setup (chain-of-thought, expert reflections), not the persona. *Request:* log and share the generation prompts and Interview 1 transcripts, so style can be tuned rather than judged and retrieval can be separated from simulation.

Table 1: Style feature means (run 1; all values pending rerun).

Feature	Human	AI
Length (words)	**[31.5]**	**[55.0]**
Lexical diversity (MTLD)	**[56.3]**	**[95.3]**
Mean word length	**[4.27]**	**[4.74]**
Contraction rate	**[0.017]**	**[0.010]**
Hedging /100w	**[5.57]**	**[2.35]**
Sentiment mean	**[0.28]**	**[0.68]**
Sentiment std	**[0.12]**	**[0.11]**
Stance /100w	**[1.70]**	**[1.42]**

Recommendation

Use the simulations for ideation, question piloting, and stress-testing survey wording — tasks that need plausible consumer-shaped text, not this consumer's truth. **Do not use** them yet for preference measurement, concept ranking, or verbatim-style insight quotes: added content and inflated positivity would systematically bias findings toward enthusiasm the real consumers do not show. The full pipeline will convert this into a per-category green/amber/red decision map.

References

Hyland, K. (2005). *Metadiscourse: Exploring Interaction in Writing*. Continuum, London. Source of the hedging lexicon.

Park, J. S., Zou, C. Q., Shaw, A., Hill, B. M., Cai, C., Morris, M. R., Willer, R., Liang, P., and Bernstein, M. S. (2024). Generative agent simulations of 1,000 people. arXiv:2411.10109; later retitled "LLM Agents Grounded in Self-Reports Enable General-Purpose Simulation of Individuals".